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Seeing Ad Transparency More Clearly: Public Perception, Use of Ad Transparency Tools, and Responses to Dataveillance

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ABSTRACT

Advertising transparency efforts aim to facilitate user knowledge and control over the use of their data for personalized advertising. We seek to understand the perceptions and use of ad transparency tools among U.S. adults and whether ad transparency information surfaced by such tools affects public perceptions of online surveillance. Results show that people are aware of, although not knowledgeable about, such tools and feel slightly satisfied with them but do not use them often. Drawing from the dataveillance effects in advertising landscape (DEAL) framework, we demonstrate that whereas exposure to no, less, or more extensive transparency information did not change perceived surveillance, exposure to more extensive targeting information increased knowledge about ad transparency efforts and self-efficacy in using transparency tools. Moreover, perceived surveillance is associated with perceived benefits and risks of personalized advertising, interest in ad transparency information, desire for privacy regulation, negative affect, and intention to protect privacy and use ad transparency tools. These relationships were moderated by privacy concern and privacy cynicism. Theoretically, this study offers an empirical test of key tenets of the DEAL framework and contributes to policy debates about ad transparency.

Technological developments in machine learning and artificial intelligence have fundamentally altered the advertising ecosystem by giving rise to digital data-driven advertising strategies, including personalized advertising that relies on the collection, mining, and aggregation of individuals' personal data (Helberger et al. 2020; Strycharz and Segijn 2022). The growth of personalized advertising has raised concerns about user privacy and "dataveillance." *Dataveillance* refers to the latent automated collection and use of digital trace data to regulate consumer behavior (Büchi, Festic, and Latzer 2022; Strycharz and Segijn 2022). Dataveillance has fueled calls for *ad transparency*, which refers to disclosures or explanations that provide information to consumers about how companies collect and use their personal data to display personalized ads (Kim, Barasz, and John 2019). Such explanations—often facilitated by ad transparency tools—help users understand how and why they were targeted for a specific ad and can include information about the advertiser, the content of the advertisement, and the targeting criteria used by the advertiser.

Many online platforms and social media networks have implemented efforts to improve ad transparency in response to public concerns about data privacy. Advertising transparency tools provide functionalities to help users understand, track, and manage advertising content on different platforms by surfacing ad transparency information to consumers (e.g., information about how and why an individual was targeted for an ad, the ad sponsor, and audience targeting criteria). Such tools can take several forms (see

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Supplementary material, Section 1, for examples of ad transparency tools). For example, publicly accessible *ad libraries* (e.g., Google and Meta) allow users to search and view active and historical ads. They typically include information about the advertiser, ad targeting criteria, ad spend, and ad impressions (i.e., ad display counts). *Ad transparency browser extensions* or *plugins* allow users to view detailed information about the ads they encounter online, such as details about the advertiser, ad targeting criteria, and tracking mechanisms. Similarly, *embedded features* on social media platforms such as the Meta "Why am I seeing this ad?" or TikTok "About this ad" buttons enable users to see how factors such as basic demographic details, interests, website visits, and matches to advertisers' targeting criteria contribute to the ads that appear on their news feed.

Although a great deal of research has investigated people's attitudes and behaviors toward personalized advertising, with a large proportion of that literature focused on privacy, less research focuses specifically on consumers' attitudes and behaviors surrounding advertising transparency. This study thus aims to shed light on public perceptions and usage of ad transparency tools. We first seek to understand U.S. adult consumers' general knowledge about, use of, and satisfaction with existing ad transparency efforts and tools. We next examine the extent to which ad transparency information manifested to consumers via an ad transparency tool affects people's dataveillance-related beliefs, attitudes about, and responses to both personalized advertising and ad transparency tools, as well as support for regulation to protect privacy from online advertisers.

This research contributes to the literature in several ways. Empirically, our study offers the first-available baseline data on public attitudes and behaviors toward ad transparency tools among U.S. adults. Studying public perceptions of ad transparency tools is important because attitudes about ad transparency tools are key to their use, and using such tools has been shown to increase knowledge about personalized advertising and help consumers make privacy decisions (Barbosa et al. 2021; Earp and Staddon 2016; Weinshel et al. 2019), influence ad effectiveness (Kim, Barasz, and John 2019), and more generally affect favorability attitudes about personalized advertising (Dogruel 2019; Samat, Acquisti, and Babcock 2017). Theoretically, our research provides an opportunity to test some central tenets of the *dataveillance in advertising landscape* (DEAL) theoretical framework (Strycharz and Segijn 2022), which has undergone limited empirical testing, and to apply this framework to the novel context of public perceptions and use of advertising transparency tools. More specifically, we test the prediction of the DEAL framework that exposure to a surveillance episode (i.e., in our case, to ad transparency information via an ad transparency tool) can shape consumers' attitudes about perceived surveillance, which then may impact consumers' cognitive, affective, and behavioral responses to personalized advertising.

From a practical standpoint, our findings about users' attitudes and behaviors offer valuable information to ad transparency tool designers. Numerous types of ad transparency efforts and tools already exist, and understanding what is deemed palatable and useful from the user perspective helps in designing tools that are tailored to users' needs. Finally, by collecting representative data our study informs policy debates about how to mitigate privacy threats posed by personalized advertising online in the U.S. context.

Advertising Transparency Tool Awareness, Use, and Satisfaction

The current status of laypeople's awareness, use, and feelings of satisfaction with the information provided by advertising transparency tools is largely unknown. Awareness of ad transparency efforts is important to study because, according to various theories including diffusion of innovations, information processing, and hierarchy of effects models (see Barry and Howard (1990) for an overview), *awareness* is the necessary first step toward adopting a new idea, practice, or product—in our case, using an ad transparency tool. However, awareness of advertising transparency tools and efforts to some extent presumes that people care about the automated collection and use of their information for personalized advertising. Ample studies confirm that many people are concerned about personalized advertising and/or feel it is a privacy violation (e.g., Auxier et al. 2019; Boerman, Kruikemeier, and Zuiderveen Borgesius 2021). Indeed, consumers' knowledge and attitudes about dataveillance in the digital data-driven advertising context are important to understand because they may impact cognitive, affective, and behavioral responses to personalized ads (Strycharz and Segijn 2022)—and, we add, to ad transparency tools themselves. For example, people who have higher concerns about being surveilled online might value and use ad transparency tools more than people with lower concerns.

Dataveillance perceptions may also motivate the use of ad transparency tools. This point is consistent

with applications of protection motivation theory and psychological reactance theory to personalized advertising (see Boerman, Kruikemeier, and Zuiderveen Borgesius 2021; Brinson, Eastin, and Cicchirillo 2018; Ham 2017; Strycharz et al. 2019). These theories suggest that self-protective behaviors, such as the use of ad transparency tools, are enacted as a means to cope with perceived privacy threats. Although researchers have studied the effect of ad transparency information on ad effectiveness (e.g., Kim, Barasz, and John 2019), for attitudes toward targeted advertising or platform trust (e.g., Dogruel 2019; Earp and Staddon 2016; Jost et al. 2023), reactions to or appreciation of ad transparency information (e.g., Balaban, Mucundorfeanu, and Naderer 2022; Eslami et al. 2018; Weinshel et al. 2019), and the limitations of ad transparency information interventions (e.g., Dobber et al. 2024), few—if any—representative studies examine the use of ad transparency tools by laypeople.

Yet understanding the use of ad transparency tools is a critical starting point to gauge their effectiveness and likely impacts on people's privacy perceptions of personalized advertising and consumer behavior. Moreover, if adopted, information about people's satisfaction with existing ad transparency tools and desire for specific ad transparency information is useful for tool developers, privacy advocates, and, potentially, to inform policymakers as they consider regulations about how online companies should provide notice to customers about the collection, use, and sharing of their personal data for advertising purposes, or whether to impose mandates to improve ad transparency (see Edelson et al. 2021).

Bearing these points in mind, we begin with an exploratory research question to investigate U.S. consumers' attitudes and behavior toward ad transparency tools. Although dataveillance is a worldwide issue, we focus on the United States for several reasons. The United States is home to some of the most popular social media platforms (e.g., Meta, X) and hosts one of the largest social media e-commerce markets globally (Statista 2023). Moreover, the adoption and use of social media are widespread across the U.S. population (Gottfried 2024), which incentivizes advertisers and service providers to engage in dataveillance. Indeed, most U.S. adults express particularly high concerns about how their data can be collected and used by institutions, and hold little faith that social media executives will protect user privacy (Faverio 2023). In addition, compared with the European Union, where regulatory frameworks such as the General Data Protection Regulation (GDPR) are well-established, the United States lacks robust privacy protection regulations at the federal level, which makes it especially important to understand user attitudes and behavior regarding ad transparency tools as one crucial step toward enhancing user privacy protection. In this context, we post our research question:

> **Research Question 1:** What is the current status of U.S. adults' (a) awareness about, (b) use of, and (c) satisfaction with available advertising transparency efforts and tools?

The Dataveillance Effects in Advertising Landscape (DEAL) Framework

The second goal of our research is to understand the effect of ad transparency information surfaced by ad transparency tools on consumers' attitudes and behaviors relevant to dataveillance. The DEAL framework (Strycharz and Segijn 2022) theorizes that users' perceptions of being surveilled by institutions, corporations, and advertisers impact their subsequent psychological and behavioral responses to such surveillance and to data-driven digital advertising more generally. Central to the DEAL framework is perceived surveillance, which encompasses feelings of being watched or monitored and that one's personal information is being recorded (Segijn, Opree, and van Ooijen 2022). According to the DEAL framework, the following are key points related to perceived surveillance: (1) consumers develop surveillance beliefs, or "the idea a person holds regarding the extent or aim of data collection" (Strycharz and Segijn 2022, 4) over time; (2) such beliefs may be formed through advertising transparency practices, advertising literacy, media coverage, and/or personal experience and folk theories; and (3) whereas surveillance beliefs are people's general awareness of surveillance practices, perceived surveillance may be activated through exposure to an observable *surveillance episode*, such as seeing a personalized ad or reading a targeting disclosure statement or explanation. The Transparency-Awareness-Control (TAC) framework (Segijn et al. 2021) similarly proposes that transparency of data collection practices is a condition for individual awareness of such practices.

We argue that exposure to ad transparency information manifested via ad transparency tools serves as a surveillance episode that may trigger or alter perceived surveillance. Prior work supports this assertion. For example, Dogruel (2019) tested the effects of exposure to advertising transparency information (operationalized by manipulating the degree of detail disclosed through an ad targeting explanation) on

consumers' evaluation of a personalized ad and trust in the social media platform where the ad appeared. Participants shown transparency information with a medium level of detail reported the most positive evaluations toward the ad, whereas people exposed to more detailed transparency information expressed less positive evaluations. Dogruel concluded that presenting users with highly detailed transparency information triggered feelings of constantly being watched by companies, which contributes to skepticism toward personalized advertising. Similarly, Kim, Barasz, and John (2019) found that ad transparency notices that reveal cross-website tracking and consumer attribute inferences as the basis for personalized advertising—practices that participants deemed highly invasive—increased participants' online privacy concerns. These results suggest that exposure to ad transparency information can affect perceived surveillance and that exposure to greater "levels" of ad transparency information—which we operationalize as the extensiveness of both the type and degree of detail of information provided by an ad transparency tool about why an ad was targeted to an individual—may increase people's perceived online surveillance. Thus, we pose the hypothesis:

Hypothesis 1: People exposed to more extensive transparency information via an ad transparency tool will report higher perceived surveillance than people exposed to less extensive or to no transparency information.

In addition to perceived surveillance, we examine beliefs and attitudes that are more directly related to policy debates about ad transparency efforts (see Edelson et al. 2021) and are useful to tool designers. Relevant questions may be asked, for example, whether exposure to different levels of ad transparency information surfaced by an ad transparency tool affects knowledge about and interest in ad transparency efforts, perceived self-efficacy in using ad transparency tools in the future, and desire for legal protection from personalized advertising. Consistent with the TAC framework (Segijn et al. 2021), it is reasonable to assume that seeing more extensive ad transparency information that potentially reveals privacy-invasive targeting practices might stimulate increased knowledge about ad transparency tools and desire for control via regulation. However, exposure to ad transparency information may have no effect on interest in or self-efficacy with ad transparency tools because people may not care about or feel powerless to prevent targeted advertising. Because theory and empirical research addressing these questions are scant, we pose the following exploratory research question:

Research Question 2: Does exposure to no transparency information versus to less extensive or to more extensive ad transparency information via an ad transparency tool affect (a) knowledge about and (b) interest in transparency information, (c) self-efficacy for using transparency tools, and (d) desire for legal protection from personalized advertising?

The Mediating Role of Perceived Surveillance

Central to the DEAL framework is that perceived surveillance, once activated by a surveillance episode (in our case, exposure to ad transparency information), may lead to surveillance responses and responses to personalized advertising (Strycharz and Segijn 2022). *Surveillance responses* include *cognitive* (e.g., privacy calculus), *affective* (e.g., emotions), and *behavioral reactions* (e.g., privacy protection behaviors). In the following discussion, we first explore how perceived surveillance may affect these three types of surveillance responses to personalized advertising. We also extend these types of responses to the context of ad transparency by arguing that feelings of being watched also impact attitudes and behavior toward ad transparency tools and also to personalized advertising more generally. We then discuss the mediating role of perceived surveillance as theorized by the DEAL framework.

Cognitive Responses

In the context of data-driven personalized advertising, cognitive responses include considerations of the costs and benefits of being surveilled by social media companies and advertisers for personalized advertising purposes (Strycharz and Segijn 2022). Indeed, Bol et al. (2018) demonstrated that people's online self-disclosure is dependent on the tradeoff between perceived risks and benefits, and that people who were shown a personalized ad reported lower trust in websites and fewer perceived benefits, likely due to heightened surveillance perceptions. Outside of the context of personalized advertising, feeling surveilled might increase people's interest in using transparency tools to learn about how their data is used and, further, their desire for legal protection (Meier, Oeldorf-Hirsch, and Krämer 2024). Yet we are aware of no large-scale studies to date that have tested the direct effect of perceived surveillance on these cognitive responses in the context of either personalized advertising or ad transparency tools. Our research thus aims to test whether perceived surveillance positively correlates with perceived risks, interest in ad transparency information, and desire for legal protection from personalized advertising and if it negatively correlates with perceived benefits of personalized advertising.

Affective Responses

Plenty of research supports that people react affectively to personalized advertising, expressing both negative and positive emotions toward it (e.g., Samat, Acquisti, and Babcock 2017; Segijn and van Ooijen 2020; Weinshel et al. 2019). In general, affective responses to data-driven personalized advertising tend to be more negative than positive (Boerman, Kruikemeier, and Zuiderveen Borgesius 2017) and reactions become more negative when personalized ads use greater amounts and more detailed user information for targeting (Boerman, Kruikemeier, and Zuiderveen Borgesius 2021; Dogruel 2019; Kim, Barasz, and John 2019; Weinshel et al. 2019). These findings suggest that greater perceptions of being surveilled online result in more negative attitudes toward personalized advertising via psychological reactance (Strycharz and Segijn 2022). Indeed, Farman, Comello, and Edwards (2020) found that perceived surveillance increased negative affective reactance toward a behaviorally targeted ad. This evidence suggests a positive correlation between perceived surveillance and negative affect toward personalized advertising.

Behavioral Responses

Behavioral responses to perceived surveillance specified by the DEAL framework include adopting privacy protection measures, limiting self-disclosure online, and refraining from online activities, among others. A central tenet of the DEAL framework is that perceptions of dataveillance can prompt actual self-protective behavior. This point is indeed theoretically substantiated by protection motivation theory and its application to online tracking and personalized advertising (e.g., Boerman, Kruikemeier, and Zuiderveen Borgesius 2021; Meier, Oeldorf-Hirsch, and Krämer 2024). First proposed by Rogers (1975), *protection motivation theory* posits that people are motivated to take action to protect themselves from perceived threats through a cognitive process that involves assessing the severity of the threat and their ability to cope with it. Such processes have been observed with regard to dataveillance online. For example, Boerman, Kruikemeier, and Zuiderveen Borgesius (2021) found that people perceive the collection and use of personal information online as a severe threat and that perceived severity positively predicted protective behavior. Meier, Oeldorf-Hirsch, and Krämer (2024) similarly found that perceiving privacy risks from ad targeting and personalization positively relates to protection motivation.

An additional behavioral response in the context of personalized advertising that has not yet been investigated is using ad transparency tools to learn how one's personal data are used for targeted advertising and protect privacy. Yet transparency information gained from using these tools may inform future privacy behavior. This result is in keeping with the Transparency-Awareness-Control (TAC) framework (Segijn et al. 2021). Applied to this context, TAC would predict that transparency about the personalization process users gain from ad transparency efforts can increase awareness about the use of personal data to generate targeted ads and perceived surveillance, which could increase the desire to control such personalization processes. That desire, when coupled with the ability to control, can prompt actual behavioral efforts to control ad personalization, including the future use of both privacy protection strategies and ad transparency tools. In other words, perceived surveillance may positively correlate with future intentions to use privacy protection strategies and ad transparency tools.

To this point, we have argued the following: (1) differences should be expected in perceived surveillance as a function of ad transparency information exposure, such that exposure to more extensive (vs. less extensive or no) ad transparency information should create higher perceived surveillance; and (2) perceived surveillance should correlate with various types of surveillance responses. This reasoning is consistent with the DEAL framework's suggestion that perceived surveillance *mediates* the effects of surveillance episodes on subsequent (Hypothesis 2) cognitive, (Hypothesis 3) affective, and (Hypothesis 4) behavioral surveillance responses (Strycharz and Segijn 2022). To test these ideas, we hypothesize that:

> **Hypotheses 2-4:** Perceived surveillance will mediate the relationships between exposure to ad transparency information and (Hypothesis 2a) perceived risks, (Hypothesis 2b) perceived benefits of personalized advertising, (Hypothesis 2c) interest in ad transparency information, (Hypothesis 2d) desire for legal protection from personalized advertising, (Hypothesis 3) negative affect toward personalized advertising, (Hypothesis 4a) future intentions to use privacy protection strategies, and (Hypothesis 4b) future intentions to use ad transparency tools.

Boundary Conditions

Finally, the DEAL framework proposes that surveillance responses may be shaped by several boundary conditions, including both individual and contextual factors. Individual differences include privacy concern, privacy cynicism, demographics, agency, persuasion and/or advertising knowledge, among others. Our study focuses on privacy concern and privacy cynicism. *Privacy concern* can be partly understood as

both cognitive and "an affective motivational disposition" that includes feelings of unease vis-à-vis the collection and processing by institutional actors such as advertisers (Dienlin, Masur, and Trepte 2023) (see Supplementary material, Section 2, for a discussion on conceptual issues of privacy concern). People with greater concern about online privacy may be more prone to feeling surveilled and thus take more action to protect themselves (Boerman, Kruikemeier, and Zuiderveen Borgesius 2017). Privacy-concerned individuals also share less, and withdraw more, personal information online (Dienlin, Masur, and Trepte 2023; Dienlin and Metzger 2016).

Privacy cynicism refers to a cognitive coping mechanism that considers privacy protection as ineffective, and can include conceptual subdimensions such as mistrust, uncertainty, powerlessness, and resignation vis-à-vis online privacy (Lutz, Hoffmann, and Ranzini 2020). People who feel cynical or powerless about the ability to stop companies from using their data may be less likely to respond to perceived surveillance (Hoffmann, Lutz, and Ranzini 2016; van Ooijen, Segijn, and Opree 2024). Lutz, Hoffmann, and Ranzini (2020) also found that increased privacy cynicism is linked to less self-protection behavior.

Following the call by Strycharz and Segijn (2022), we thus study privacy concern and privacy cynicism as boundary conditions for how people respond to feeling surveilled. In particular, as both pertain to an underlying lack of control over personal information, high privacy concern and low privacy cynicism may strengthen the relationship between perceived surveillance and the surveillance responses in the mediation models proposed above. We thus hypothesize:

> **Hypothesis 5:** Privacy concern and privacy cynicism will moderate the relationships hypothesized in Hypotheses 2-4, such that the predicted effects of perceived surveillance on surveillance responses in Hypotheses 2-4 will be stronger among those who report (a) higher privacy concern and (b) lower privacy cynicism.

Figure 1 presents our hypothesized conceptual model.

Method

This research was approved by the institutional review board of UC Santa Barbara (protocol number: 46-24-0144). To test our predictions we employed a survey experiment. Survey experiments are common in the social sciences and offer the advantage of providing both generalizability of the findings and the ability to test causality (Auspurg and Hinz 2014; Diaz, Grady, and Kuklinski 2020; Schnabel 2022). A *survey experiment* is simply an experiment conducted within a survey. We employed this method because it allowed us to achieve our two research goals. For our first goal of understanding public perceptions of ad transparency tools and personalized advertising in the population, we needed to survey a representative sample of U.S. adults. Our second goal was to understand if exposure to different levels (i.e., extensiveness) of ad transparency information, as manifested to consumers via a prototypical ad transparency tool, changes public perceptions, which required experimental control.

In our study, while all participants completed a survey about ad transparency and personalized advertising, a subset of the total participants were randomly assigned to additionally see one of two examples of information surfaced by an ad transparency tool that was embedded in the survey. This approach enabled us to test if exposure to a surveillance episode in the form of an explanation about ad targeting delivered via an ad transparency tool impacts people's surveillance responses. Similar to Dogruel (2019), we tested two different explanations to understand the extent to which different types of information and degrees of detail in the transparency information manifested by an ad transparency tool impact public attitudes. For this purpose, we randomly assigned 300 participants to see either a "less extensive" or a "more extensive" ad transparency information version of the experimental stimuli. The remaining participants constituted the "no exposure" control group. These participants completed the survey but did not view either version of the stimuli. The control group was larger in order to provide baseline data representative of U.S. adults to answer our first set of research questions, and to generalize our findings about ad transparency tools and personalized advertising to the U.S. adult population. See Table 1 for sample characteristics.

The data were collected on Prolific in March, 2024 from a total of 1,352 participants. The data were cleaned to eliminate the following participants: speeders (anyone who completed the study in less than one-third of the median completion time; $n = 15$); those who gave intentionally low-effort responses (i.e., straight-lining, identified by examining the standard deviations of participants' responses and via visual inspection of each participant's responses in the dataset; $n = 189$), and those who failed the attention check ($n = 3$) and/or manipulation check ($n = 13$). After data cleaning, a total of 1,132 participants remained, with 137 participants in each of the two treatment groups and 858 participants in the baseline control group. The median time for participants to complete the study was between 12 and 13 minutes.

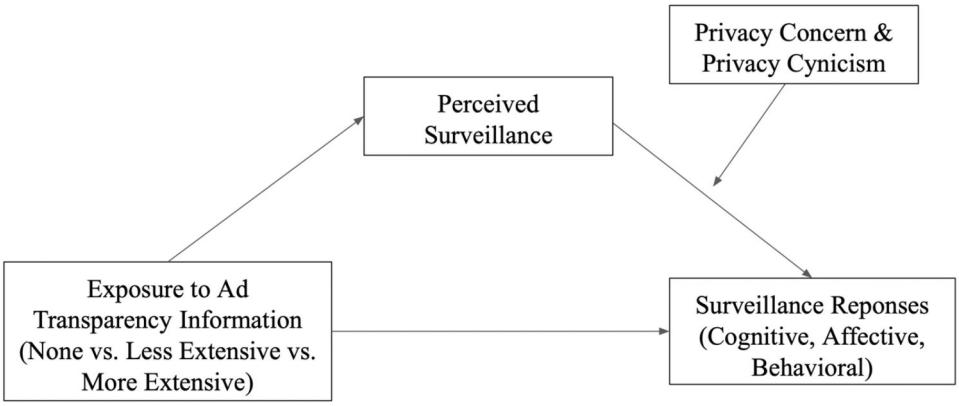


Figure 1. Hypothesized moderated mediation model based on the DEAL framework.

Table 1. Sample descriptives for baseline control group ($N = 858$).

Characteristic	Sample descriptives
Age	Mean: 47 years Standard deviation: 16.86 years
Gender/sex identity	51.2% Female ($n = 439$) 45.7% Male ($n = 392$) 3.0% Non-binary or other ($n = 26$)
Income	Mean: Between $50,000 and $74,999 Range: < $25,000 to ≥$150,000
Education	Median: Bachelor's degree Range: Less than a high school degree to post-graduate degree
Race and ethnicity	14.0% Black, African, or African American ($n = 120$) 11.9% Hispanic, Latina/Latino, Latine, or Latinx ($n = 102$) 69.0% White ($n = 592$) 7.0% Asian ($n = 60$) .2% Native Hawaiian or Other Pacific Islander ($n = 2$) 1.7% Native American, American Indian, or Alaskan Native ($n = 15$) 2.0% Other ($n = 17$)
Political ideology	Mean: 3.36 (moderate to slightly liberal)
Time spent on social media per day	Mean: between "most days" and "every day"

Note. The study sample as a whole ($N = 1,132$) was also representative of the U.S. adult population.

Procedure and Stimuli

Figure 2 displays a visual overview of our procedure. Upon consent, all participants read a definition of the term "advertising transparency" (i.e., information about ad sponsors and who they are targeting) and a brief description of current ad transparency tools, platforms, and efforts to orient them to the study. They were then asked about their awareness of ad transparency as a concept prior to the study, as well as their previous use of and satisfaction with any existing ad transparency tools, knowledge about ad transparency, perceived surveillance, surveillance responses, intended future use of ad transparency tools, as well as other variables suggested by the DEAL framework.

All participants completed the same instrument, and a subset of participants additionally viewed one of two experimental stimuli, as previously described. The stimuli were inserted immediately after the definition of advertising transparency and the questions about prior awareness of, use of, and satisfaction with available ad transparency tools. The stimuli depicted the results of a typical ad transparency tool of the "Why am I seeing this ad?" variety that is commonly used on the most popular U.S. social media platforms. Participants in the less and more extensive ad transparency information conditions were told, "Below is a screenshot of an ad that was personalized for an Instagram user, with ad transparency information shown in the photo on the right." In the left photo, the screenshot displayed an ad for a food delivery service with "Why you're seeing this ad" overlayed on it. The ad targeting explanation was displayed in the right photo (see Figures 3 and 4). Our

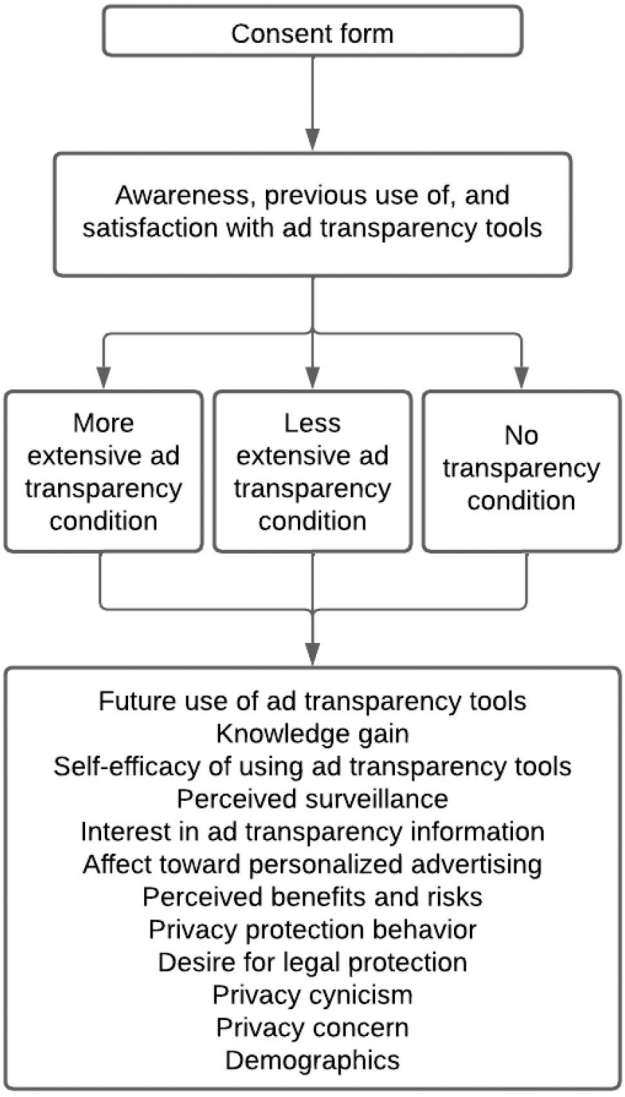


Figure 2. Flowchart showing overview of study procedure.

study design was modeled on previous research (e.g., Dogruel 2019; Kim, Barasz, and John 2019; Meier, Oeldorf-Hirsch, and Krämer 2024) that employed similar manipulations.

The ad explanation in the more extensive transparency information condition displayed a greater range of and more detailed user information that the targeted ad was purportedly based on compared to the explanation displayed in the less extensive ad transparency information condition. Specifically, the less extensive transparency information condition indicated targeting was based only on the users' broad age range (older than 18 years) and country-level location (see Figure 2). The more extensive transparency information condition informed participants that the ad was targeted to the user based on the following criteria: a narrower age range (18–24 years); location; inferred interests; language; video viewing history; device information; activities on and off the platform; and a catch-all "other" user data or ad sponsor targeting criteria category as the basis for seeing the ad (see Figure 3). Participants in the control condition saw no ad transparency tool or transparency information. Participants in the treatment conditions then answered the same questions about their future use of ad transparency tools, knowledge about ad transparency, perceived surveillance, surveillance responses, as well as other variables, as did participants in the control group.

The social media platform, the ad, and the ad transparency information used in the stimuli were chosen to ensure ecological validity. Instagram was selected because it is currently the most popular social media app in United States An ad for a food delivery service was chosen because it is applicable to most people compared with ads for many other products or services. Our stimulus for the less extensive transparency information condition was modeled on current targeting explanations shown by typical "Why am I seeing this ad?" and "About this Ad" tools in major U.S. social media platforms. To explore user reactions to potential ad explanations that are more comprehensive and that also integrate design suggestions from prior work (e.g., Eslami et al. 2018; Wei et al. 2020), the tool shown in the "high" transparency information condition included both more, and more detailed, types of ad targeting information, which were in fact real ad transparency information that one could surface on social media using currently available ad transparency tools.

Experiment Manipulation Check

To check the ad transparency information manipulation, participants exposed to the ad transparency stimuli were asked to recall the types of information that were used to personalize the ad for the user from a list that was provided, including the following user data: age; primary location; estimate of the user's interests; language; video viewing history; device type; activities on or off Instagram; and a catch-all additional information option. Participants in the less extensive ad transparency information condition recalled seeing between two and three types of information ($M = 2.43$, SD $= .82$), which was significantly less transparency information than what was reported by those in the more extensive ad transparency information condition ($M = 7.43$, SD $= 1.27$), $t(293.89) = -43.71$, $p < .001$, Cohen's $d = 1.07$). This finding indicates that the manipulation was successful. Only 4% ($n = 7$) of participants in the less extensive and 6.7% ($n = 12$) in the more extensive transparency

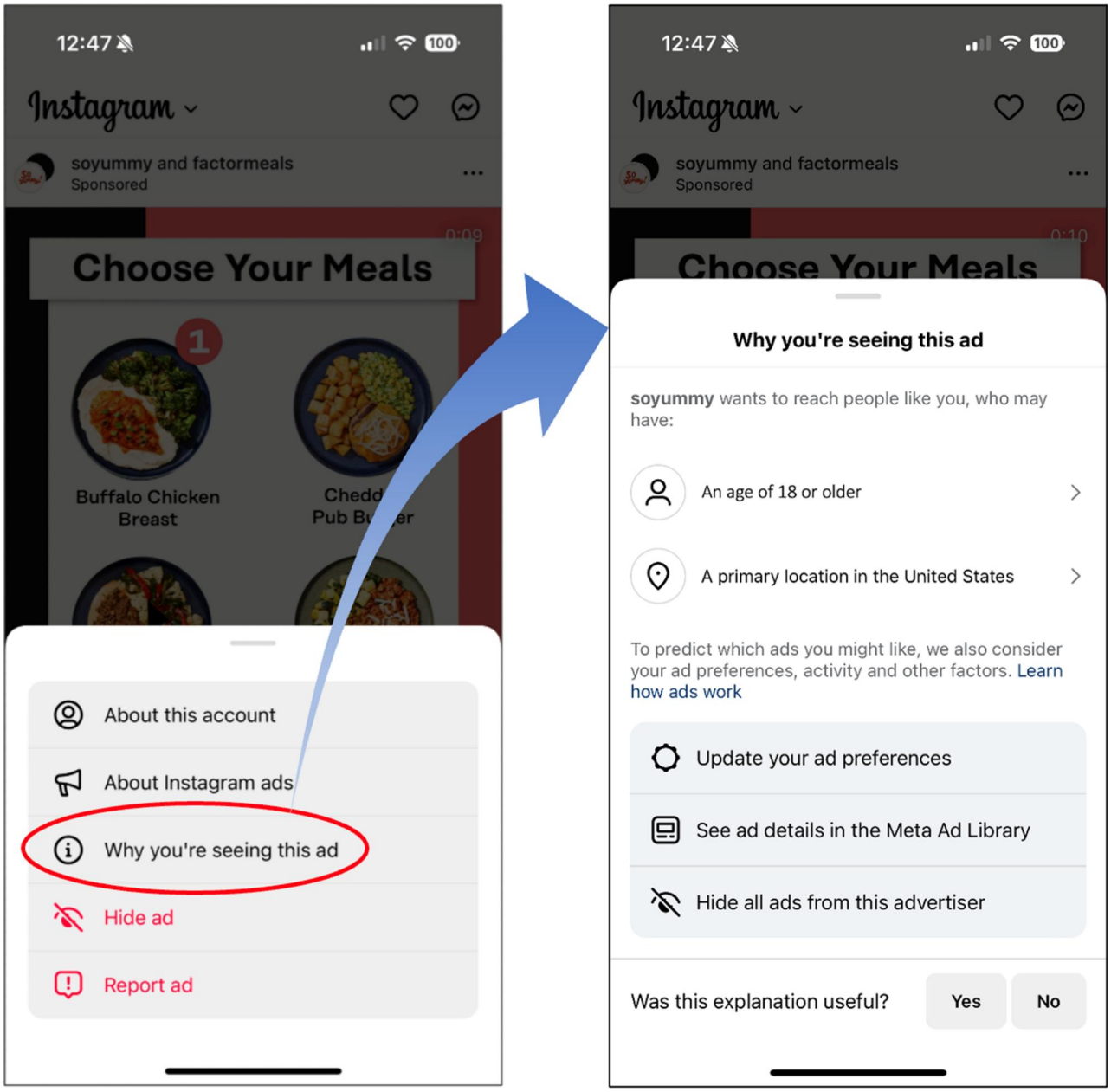


Figure 3. Less extensive transparency information manipulation.

information conditions failed the manipulation check in terms of misidentifying the types of transparency information they were shown. These participants were removed prior to the analyses.

Measures

Upon reading the description of advertising transparency at the outset of the study, *advertising transparency awareness* was measured by having participants indicate how much they had heard about any relevant efforts or tools prior to the study on a 5-point scale (1 = "I'd never heard about ad transparency before," 5 = "I knew a lot about ad transparency and I feel pretty knowledgeable about it").

Participants reported their *previous use of advertising transparency tools* by indicating how often they had used various ad transparency tools or platforms in the past on a 5-point scale (1 = Never, 5 = All the time). They were presented with a battery of 14 possible uses (e.g., "Explore an ad archive to view past ads that were shown to you," and "Find information about an advertiser who targeted you with an ad").

Satisfaction with the information provided by advertising transparency tools was gauged by having participants indicate how satisfied they were with the

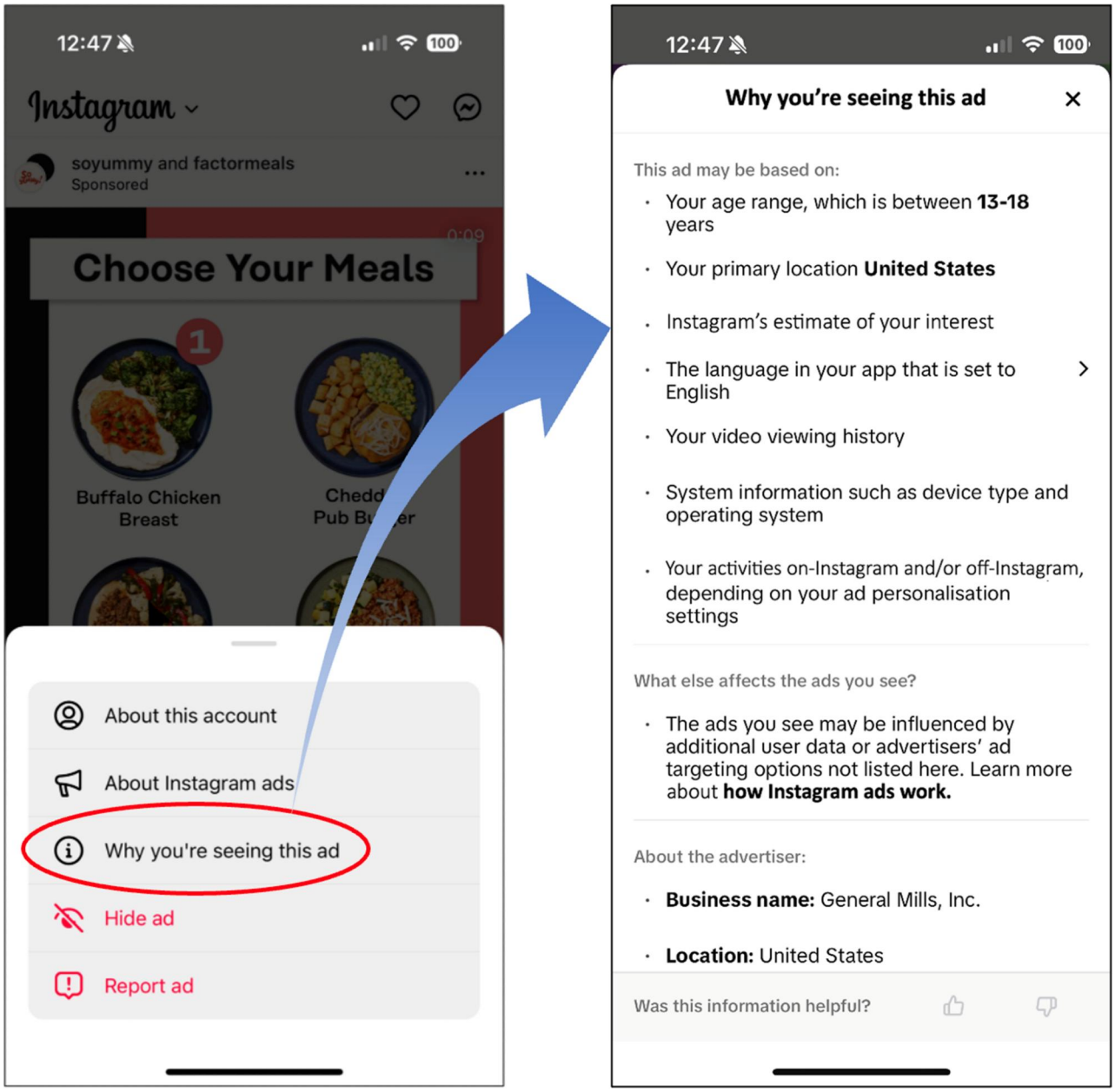


Figure 4. More extensive transparency information manipulation.

information they were able to get from ad transparency tools in the past on a 7-point scale (1 = Extremely dissatisfied, 7 = Extremely satisfied). The instrument was set up such that participants who reported they had never used an ad transparency tool in the past did not answer this question.

To measure *future use of advertising transparency tools* participants indicated how interested they were in using ad transparency tools in the future to find out why they were targeted for a specific ad or to learn more about the advertiser on a 7-point scale (1 = Not at all interested, 7 = Extremely interested).

For *knowledge gain*, participants indicated how much they agreed with the statement, "I feel I know more about ad transparency and/or ad transparency tools than I did before taking this survey" on a 7-point scale (1 = Strongly disagree, 7 = Strongly agree).

Inspired by Boerman, Kruikemeier, and Zuiderveen Borgesius (2021), we used two items to measure *self-efficacy for using ad transparency tools*, including, "I feel I am now better able to use ad transparency tools in the future" and "I feel more confident in using ad transparency tools in the future."

Adopting Segijn, Opree, and van Ooijen (2022), *perceived surveillance* was measured on a 7-point scale (1 = Not at all, 7 = Very much) by asking participants the extent to which they feel that advertisers on social media or the Internet are "Looking over your shoulder," "Watching your every move," "Entering your private space," and "Checking up on you."

Interest in advertising transparency information was measured by having participants report how interested they were in knowing 13 specific types of ad transparency information, such as "How your data is collected and used for personalized advertising purposes" and "Audiences that a particular advertiser is trying to reach" on a 7-point scale (1 = Not at all interested, 7 = Extremely interested).

The *desire for legal protection* measure draws from Meier, Oeldorf-Hirsch, and Krämer (2024). Participants indicated how much they agree with four statements about how lawmakers should better handle personalized advertisements on the Internet, such as "I think it would be good if the law placed more emphasis on protecting users from targeted advertising" on a 7-point scale (1 = Strongly disagree, 7 = Strongly agree).

Participants indicated their *affect toward personalized advertising* on a 7-point bipolar scale that included the adjectives: "Good/Bad, Negative/Positive, Favorable/Unfavorable, Dislikable/Likable" following Cotte, Coulter, and Moore (2005) and Homer (1990), with higher scores indicating a greater negative affect.

Perceived benefits and risks of personalized advertising were measured by asking participants to indicate how much they agree they benefit or could be harmed from personalized ads for different reasons on a 7-point scale (1 = Strongly disagree, 7 = Strongly agree). Seven items measured benefits (e.g., Personalized ads "are more informative to me," "reduce the number of total ads or irrelevant ads that I see") and eight items measured risks (e.g., Personalized ads "are creepy or make me feel uncomfortable," "use my personal data to try to manipulate me"). Both scales were inspired by existing measures (e.g., van Ooijen, Segijn, and Opree 2024) and qualitative research on personalized advertising (e.g., Segijn and van Ooijen 2022).

For *privacy protection behavior*, participants indicated their likelihood of engaging in seven privacy protection strategies adapted from Boerman, Kruikemeier, and Zuiderveen Borgesius (2021) on a 7-point scale (1 = Extremely unlikely, 7 = Extremely likely), such as "Turn off personalized advertising in my social media settings" and "Report and hide targeted ads in order to see less of them."

General *online privacy concern* was measured by adopting items from existing research (Dienlin and Metzger 2016; Malhotra, Kim, and Agarwal 2004; see also Smith, Milberg, and Burke 1996). Participants were asked to indicate how much they agree on a 7-point scale (1 = Strongly disagree, 7 = Strongly agree) with three statements: "Before taking this survey today, would you say that: (1) "I am a person who worries about my privacy as a result of using the Internet;" (2) "I am a person who does not feel especially concerned about my privacy when I go online (reverse-coded); and (3) "I am a person who feels the danger to people's privacy when they use the Internet has been overblown" (reverse-coded). An analysis of variance showed that membership in the treatment and control groups did not affect this trait-level measure ($F(2, 1129) = .574$, $p = .56$).

Privacy cynicism was measured by adapting the powerlessness subdimension of the privacy cynicism scale by Lutz, Hoffmann, and Ranzini (2020). On a 7-point scale (1 = Strongly disagree, 7 = Strongly agree), participants indicated how much they agreed with five statements (e.g., "Even if I try to protect my data, I can't prevent online advertisers from accessing them").

Participants also reported their *demographics*, including age, educational attainment, income, racial/ethnic identity, sex/gender identity, and political affiliation. A complete list of measurement items and information about their reliability and validity are available in the Supplementary material, Sections 3 and 4.

Analytical Strategy

Research Question 1, which sought to understand the current status of U.S. adults' attitudes toward and use of ad transparency tools, was answered by inspecting descriptive statistics (means and standard deviations). Because these questions were asked of all participants prior to the stimuli being presented to the treatment groups, responses from all study participants were used in these analyses. Hypothesis 1 and Research Question 2 asked about differences in perceived surveillance, knowledge about and interest in ad transparency information, self-efficacy, and desire for legal protection between the ad transparency conditions. These points were tested using analysis of covariance controlling for demographics (i.e., age, sex, education) and experience with transparency tools (i.e., prior awareness of ad transparency information, previous use of ad transparency tools, and satisfaction with ad transparency information).

Hypotheses 2-4 hypothesized that perceived surveillance would mediate the effects of exposure to ad transparency information on cognitive, affective, and behavioral outcomes. Mediation analyses were conducted using PROCESS (Model 4) with several covariates (i.e., age, sex, education, awareness of ad transparency information) (reference group = no ad transparency information). Hypothesis 5 predicted moderating effects of (a) privacy concern and (b) privacy cynicism for the impact of perceived surveillance on surveillance responses. Moderated mediation analyses were conducted using PROCESS (Model 14) in which privacy concern and privacy cynicism moderated the relationships between perceived surveillance and surveillance responses in the mediation models above. The same set of covariates (i.e., age, sex, education, awareness of ad transparency information) were included. The continuous predictor and moderator variables were mean-centered.

Results

Awareness, Use, and Satisfaction

Research Question 1 asked about the current status of awareness about, use of, and satisfaction with advertising transparency efforts and tools among the adult U.S. population. Inspections of the means and standard deviations showed that the average awareness score fell between "I'd heard of the idea of getting information about online ads, but don't know anything beyond that" and "I knew what ad transparency is, but I don't have a lot of knowledge about how it works" ($M = 2.60$, SD $= 1.03$, range $=$ 1–5), and 17.9% participants reported having never heard about ad transparency before. As shown in Table 2, participants reported rarely to sometimes using ad transparency tools ($M = 2.06$, SD $=$ .72, range $=$ 1–5). Ad transparency tools participants used the most frequently include opting-out of personalized advertising, ad preferences pages, and "Why Am I Seeing this Ad?/About this Ad" buttons. Participants' satisfaction with ad transparency tools was between "neither satisfied nor dissatisfied" and "slightly satisfied" ($M = 4.25$, SD $= 1.29$, range $=$ 1–7).

Table 2. Descriptives for ad transparency tool use ($N = 1{,}132$).

Item	*M*	SD
Explore or scroll through an ad transparency website or platform (e.g., Google or Meta's Ad Transparency Center or Ad Library)	1.62	.99
View your "ad preferences," which is the information that advertisers use to target you (e.g., your interest topics, demographics) to learn the reasons why you got a particular ad you saw online	2.35	1.13
Explore an ad archive to view past ads that were shown to you	1.47	.86
Change or adjust your ad preferences (e.g., to be shown fewer ads on a specific topic)	2.74	1.23
Change your personal information on your profile to avoid targeted advertising	2.21	1.23
Opt out/turn off personalized ads	3.08	1.22
Learn how data about you was collected and used for personalized advertising purposes	2.15	1.07
Find information about an advertiser who targeted you with an ad	1.98	1.06
Click on a "Why am I seeing this ad?" or "About this Ad" button for a particular ad	2.60	1.11
Learn about general ad policies, guidelines, and practices	2.05	1.05
Access user education resources about online advertising	1.62	.95
Learn about the kinds of audiences that an advertiser is trying to reach	1.83	1.04
Find out the amount of money the advertiser spent on the ad	1.42	.85
Discover who actually paid for an ad that you saw	1.68	.97
Average across all items	2.06	.72

Note. Participants were asked how often they had used an ad transparency tool to do each of the activities listed in the table. Scores ranged from 1 (never) to 5 (all the time).

The Impact of Ad Transparency Information

Hypothesis 1 predicted that people exposed to more extensive transparency information via an ad transparency tool would report higher perceived surveillance online than those exposed to less extensive or to no transparency information. The Levene test indicated a violation of the homogeneity of variance assumption, so we obtained parameter estimates with robust standard errors in addition to the F-test (see Supplementary material, Section 5 for more information). Results showed no significant differences in perceived surveillance among people exposed to the three different "levels" of transparency information we tested ($F(2, 1037) = 1.29$, $p = .28$). The t-tests of the heteroskedasticity-consistent standard errors using the heteroskedasticity-consistent 3 (HC3) method were also nonsignificant, $t_{(condition\ =\ 0)} = -1.36$, $p = .17$ and $t_{(condition\ =\ 1)} = -1.51$, $p = .13$. Although, descriptively, people exposed to the most extensive transparency information ($M = 4.91$, SD $= 1.44$) reported higher perceived surveillance than people exposed to less extensive transparency information ($M = 4.61$, SD $= 1.73$), and to the no-transparency information control group ($M = 4.82$, SD $= 1.58$), the differences did not reach statistical significance, as noted previously. Hypothesis 1 is not supported.

Research Question 2 asked about differences in (a) knowledge about and (b) interest in ad transparency, (c) self-efficacy for using transparency tools, and (d) desire for legal protection from personalized advertising among people exposed to varying levels of ad transparency information. Significant differences were found in

knowledge about ad transparency ($F(2, 1037) = 11.26$, $p < .001$, $\eta_p^2 = .02$). Post-hoc tests using Bonferroni corrections showed that people exposed to more extensive transparency information reported greater knowledge about ad transparency and transparency tools ($M = 5.47$, SD $= 1.25$) than people exposed to no transparency information ($M = 5.02$, SD $= 1.32$, $p < .001$), but scores in the more extensive transparency information condition did not differ from the less extensive transparency information condition ($M = 5.32$, SD $= 1.31$, $p = .66$). The extensiveness of the transparency information similarly influenced people's reported self-efficacy for using transparency tools ($F(2, 1037) = 20.48$, $p < .001$, $\eta_p^2 = .04$). People exposed to more extensive transparency information ($M = 5.27$, SD $= 1.11$) or to less extensive transparency information ($M = 5.20$, SD $= 1.00$) reported greater self-efficacy for using transparency tools than people exposed to no transparency information ($M = 4.80$, SD $= 1.16$, $p < .001$), although being exposed to more extensive information did not lead to greater perceived self-efficacy than being exposed to less extensive information ($p = 1.00$). No significant differences were found, however, for interest in ad transparency tools ($F(2, 1037) = .074$, $p = .93$) or desire for legal protection by regulation ($F(2, 1037) = 2.25$, $p = .11$).

Perceived Surveillance and Surveillance Responses

Hypotheses 2-4 posited that perceived surveillance would mediate the effects of ad transparency exposure on (Hypothesis 2a) perceived risks, (Hypothesis 2b) perceived benefits of personalized advertising, (Hypothesis 2c) interest in ad transparency information, (Hypothesis 2d) desire for legal protection, (Hypothesis 3) negative affect toward personalized advertising, (Hypothesis 4a) future intentions to use privacy protection strategies, and (Hypothesis 4b) future use of ad transparency tools. Consistent with findings for Hypothesis 1, after controlling for age, sex, education, and ad transparency awareness, perceived surveillance did not differ as a function of exposure to ad transparency information across all mediation models. The hypothesized mediation relationships in Hypotheses 2-4 were therefore not significant (see Supplementary material, Section 6, for more information about the indirect effects). In other words, exposure to ad transparency did not impact surveillance responses indirectly through perceived surveillance.

After controlling for ad transparency information exposure, however, results indicate direct relationships between perceived surveillance and all types of surveillance responses, which aligns with the DEAL propositions, (see Table 3). In particular, perceived surveillance positively predicted perceived risks of personalized advertising, interest in ad transparency information, desire for legal protection, negative affect toward personalized advertising, and future intentions to use privacy protection strategies and ad transparency tools. Perceived surveillance also negatively predicted perceived benefits of personalized advertising.

Boundary Conditions

Hypothesis 5 predicted that privacy concern and privacy cynicism would moderate the hypothesized mediation relationships. Specifically, the positive effect of perceived surveillance on surveillance responses should be stronger among those who report (a) higher privacy concern, and (b) lower privacy cynicism. The coefficients for the direct effects and simple slopes for the interactions can be found in Tables 4 and 5. Figures showing the significant moderations and conditional indirect effects with 95% confidence intervals can be found in the Supplementary material, Sections 6 and 7.

For Hypothesis 5a, moderation analyses revealed a negative conditional main effect for privacy concern on perceived benefits, and positive conditional main effects for perceived risks, interest in ad transparency information, desire for legal regulation, negative affect,

Table 3. Direct effects of mediations (Hypothesis 2, Hypothesis 3, and Hypothesis 4).

Hypotheses		Less transparency (vs. control)	More transparency (vs. control)	Perceived surveillance
H2a	Mediator: Perceived surveillance	−0.17	.14	–
	Outcome: Perceived risks	−0.14	.07	.31***
H2b	Mediator: Perceived surveillance	−0.17	.14	–
	Outcome: Perceived benefits	.07	−0.11	−0.07*
H2c	Mediator: Perceived surveillance	−0.17	.14	–
	Outcome: Interest	.00	−0.03	.22***
H2d	Mediator: Perceived surveillance	−0.17	.14	–
	Outcome: Desire	−0.20	−0.10	.23***
H3	Mediator: Perceived surveillance	−0.17	.14	–
	Outcome: Negative affect	.24	−0.19	.14***
H4a	Mediator: Perceived surveillance	−0.17	.14	–
	Outcome: Privacy protection	.17	−0.02	.26***
H4b	Mediator: Perceived surveillance	−0.17	.14	–
	Outcome: Future use	−0.02	−0.18	.26***

Note: *$p < .05$, **$p < .01$, ***$p < .001$.

Table 4. Direct and conditional effects of moderated mediations (moderator: privacy concern) (Hypothesis 5).

Dependent variable	Independent variable Less transparency (vs. control)	Independent variable More transparency (vs. control)	Independent variable Perceived surveillance	Moderator Privacy concern	Interaction term	Conditional effects		
						Low privacy concerns (−1 SD)	Average privacy concerns	High privacy concerns (+1 SD)
Perceived risks	−0.13	.07	.25***	.26***	−0.04*	.30***	.25***	.21***
Perceived benefits	.05	−0.08	−0.02	−0.34***	−0.06**	.05	−0.02	−0.08*
Interest in ad transparency information	.01	−0.03	.20***	.10***	−0.02	–	–	–
Desire for legal protection	−0.18	−0.10	.15***	.40***	−0.03*	.19***	.15***	.11***
Negative affect	−0.22	.16	.07*	.46***	.06**	−0.00	.07*	.13***
Future use of privacy protection strategies	.19	−0.02	.19***	.36***	−0.04*	.24***	.19***	.14***
Future use of ad transparency tools	−0.01	−0.12	.23***	.11**	−0.05*	.28***	.23***	.17***

Note. Coefficient estimates are unstandardized. The independent and moderating variables are mean centered. $p < .05^{*}$, $p < .01^{**}$, $p < .001^{***}$.

Table 5. Direct and conditional effects of moderated mediations (moderator: privacy cynicism) (Hypothesis 5).

Dependent variable	Independent variable Less transparency (vs. control)	Independent variable More transparency (vs. control)	Mediator Perceived surveillance	Moderator Privacy cynicism	Interaction term	Conditional effects		
						Low privacy cynicism (−1 SD)	Average privacy cynicism	High privacy cynicism (+ 1 SD)
Perceived risks	−0.16	.08	.28***	.14***	.00	–	–	–
Perceived benefits	.09	−0.12	−0.04	−0.17***	−0.00	–	–	–
Interest in ad transparency information	.01	−0.03	.23***	−0.04	.00	–	–	–
Desire for legal protection	−0.22*	−0.10	.21***	.10***	.02	–	–	–
Negative affect	−0.27*	.20	.10***	.21***	.03	–	–	–
Future use of privacy protection strategies	.17	−0.02	.26***	.01	.01	–	–	–
Future use of ad transparency tools	−0.01	−0.17	.27***	−0.06	.05*	.21***	.27***	.33***

Note. Coefficient estimates are unstandardized. The independent and moderating variables are mean centered. $p < .05^{*}$, $p < .01^{**}$, $p < .001^{***}$.

use of privacy protection strategies, and use of ad transparency tools, while controlling for ad transparency information exposure, perceived surveillance, the interaction term of perceived surveillance and privacy concern, and the covariates (i.e., age, sex, education, ad transparency awareness, experimental condition). For Hypothesis 5b, a negative conditional main effect for privacy cynicism was found on perceived benefits, and positive conditional main effects were found on perceived risks, desire for legal regulation, negative affect, and future use of ad transparency tools.

For both Hypotheses 5a and 5b, however, results showed that the moderated mediations were not statistically significant; therefore, the overall mediation models were not contingent upon the levels of the moderators. However, we found that privacy concern and privacy cynicism moderated the direct relationships between perceived surveillance and several surveillance responses, as discussed in following text.

Cognitive Responses

Privacy concern moderated the impact of perceived surveillance on *perceived risks of personalized advertising*. Simple slopes analysis showed the conditional effect of perceived surveillance on perceived risks was positive and significant across all levels of privacy concern (low, average, high), although the effect was stronger among less concerned individuals. Privacy cynicism did not moderate the relationship between perceived surveillance and perceived risks.

Privacy concern moderated the impact of perceived surveillance on *perceived benefits*, such that the conditional effect of perceived surveillance on perceived benefits was significant only when privacy concern was high. In other words, perceived surveillance negatively predicted perceived benefits only among highly concerned individuals. However, privacy cynicism did not moderate the relationship between perceived surveillance and perceived benefits.

Neither privacy concern nor privacy cynicism significantly moderated the relationship between perceived surveillance and *interest in ad transparency information*. Thus, regardless of how concerned or cynical people feel about online privacy, the more they feel being surveilled, the more interest they show in ad transparency information.

Privacy concern moderated the relationship between perceived surveillance and *desire for legal*

protection from personalized advertising, such that the conditional effect of perceived surveillance on desire for legal protection was positive and significant across all levels of privacy concern, although the effect was stronger among less concerned individuals. Privacy cynicism did not moderate this relationship.

Affective Responses

Privacy concern moderated the relationship between perceived surveillance and *negative affect*, such that the conditional effect of perceived surveillance on negative affect was positive and significant only among those who reported average and high privacy concern. The positive effect of perceived surveillance on negative affect was especially strong among highly concerned individuals. Privacy cynicism was not a significant moderator.

Behavioral Responses

Privacy concern moderated the relationship between perceived surveillance and the *use of privacy protection strategies*, such that the conditional effect of perceived surveillance on the use of strategies was positive and significant across all levels of privacy concern, although the effect was stronger among less concerned individuals. Privacy cynicism did not moderate this relationship.

Privacy concern moderated the relationship between perceived surveillance and *future use of ad transparency tools*, such that the conditional effect of perceived surveillance on future use of ad transparency tools was positive and significant across all levels of privacy concern, although the effect was stronger among less concerned individuals. The moderating role of privacy cynicism was also significant. The conditional effect of perceived surveillance on future use of ad transparency tools was positive and significant across all levels of privacy cynicism, although the effect was stronger among more cynical individuals.

Discussion

This research set out to understand the public perceptions and use of ad transparency tools among U.S. adults as a potential solution to privacy threats posed by dataveillance practices used in personalized advertising, to investigate how ad transparency information provided by such tools impact attitudes about and outcomes associated with dataveillance, and to offer an empirical test of key tenets of the DEAL framework in terms of responses to personalized advertising and ad transparency tools. Drawing from a representative survey of the U.S. population with an embedded experiment, our results contribute to the growing literature on and recent theorizing of how current ad transparency efforts and perceived surveillance impact responses to personalized advertising.

Ad Transparency Tool Perceptions and Usage

Our first set of findings revealed that, whereas most of our survey respondents reported having heard of ad transparency efforts and tools, most also said they have little knowledge about them beyond this. Moreover, almost one-fifth of respondents said they had no familiarity with these concepts at all. These results are consistent with Eslami et al. (2018). Many (31.4%–88.1%) also reported having never or rarely used any of the various specific ad transparency tools we asked about. The data further show that ad transparency tools are used the least to find out an ad sponsor's identity, ad spend, and general targeting criteria and to view ad archives or transparency education centers or ad libraries, whereas transparency tools are used the most to opt out of personalized ads, adjust ad preferences, and see why a particular ad targeted a user, although these tools were still used for these purposes only rarely to sometimes. Among those who had used at least one type of ad transparency tools, the individuals were not highly satisfied with the information that the tool(s) provided.

These results indicate a need for more effort to increase public awareness of the concept of ad transparency, as well as available tools and options for users to understand how advertisers conduct digital surveillance. This need is particularly pressing for individuals and groups who may have fewer digital resources yet are more likely to be algorithmically discriminated against in personalized advertising scenarios (Ali et al. 2019). The results also indicate that such efforts should specifically include the full range of data that can be surfaced by available ad transparency tools. Our research speaks to what information and capabilities consumers tend to seek out when using available ad transparency tools, which has important implications for policymakers and tool designers. For example, we found that people particularly appreciate opt-out tools that allow them to stop advertisers from targeting them and tools to help them understand the targeting criteria advertisers use to show them personalized ads.

Perceived Surveillance

Consistent with previous research (Segijn, Opree, and van Ooijen 2022; Segijn et al. 2021; Sifaoui, Lee, and Segijn 2023), our data show that Americans do indeed feel like they are being watched by advertisers when they are online, as demonstrated by the average scores being above the midpoint on our perceived surveillance measure (see Supplementary material, Section 8 for more information about our findings on perceived surveillance). However, while we argued that the extensiveness of ad transparency information provided would impact the extent to which people feel surveilled online, we found no significant difference in perceived surveillance among those exposed to more extensive, less extensive, or no ad transparency information in our experiment. This result is consistent with findings that more extensive ad explanations did not increase perceived privacy risks (Meier, Oeldorf-Hirsch, and Krämer 2024) or privacy concerns (Dobber et al. 2024), although those studies were in a political advertisement context.

Conversely, other studies have found more extensive transparency information lowered ad evaluations due to priming feelings of being watched (Eslami et al. 2018) or that the ad is a privacy violation (Kim, Barasz, and John 2019; see also Dogruel (2019)). One explanation for why our manipulations did not produce significant differences in perceived surveillance could be that participants came into our study with a high baseline level of perceived surveillance, producing a ceiling effect. Indeed, while the more (vs. less) extensive transparency information condition elicited greater surveillance perceptions, neither was significantly different from the no-transparency information condition and all group means were above the scale midpoint. Therefore, although our manipulations worked, providing participants with more or less extensive ad transparency information apparently did not change their previously settled perceptions of advertiser surveillance (see also Meier, Oeldorf-Hirsch, and Krämer 2024). In any case, this result is perhaps good news to ad transparency advocates because our finding helps alleviate the concern that exposure to ad transparency information may result in heightened privacy concerns and potential chilling effects (Büchi, Festic, and Latzer 2022; Dobber et al. 2024).

Adding to this optimistic perspective, both higher knowledge about ad transparency and self-efficacy for using transparency tools were observed among those exposed to ad transparency information in our experiment compared with those not exposed to such information. These findings are encouraging to tool designers and policymakers because seeing transparency information likely increases user awareness and motivates future use of such tools.

Surveillance Responses

A central tenet of the DEAL framework is that perceived surveillance acts as an underlying mechanism for cognitive, affective, and behavioral surveillance responses. Our results did not support the mediating role of perceived surveillance as implied in the DEAL framework, likely due to the fact that, again, perceived surveillance did not differ as a function of ad transparency exposure. However, consistent with the DEAL proposition, we found that perceived surveillance positively predicted perceived risks of personalized advertising, interest in transparency information, desire for legal regulation of personalized advertising, negative affect toward personalized advertising, and future intentions to use privacy protection strategies and ad transparency tools, as well as negatively predicted perceived benefits of personalized advertising.

These results contribute to the growing literature on and theorizing of dataveillance effects by shedding light on the important role of perceived surveillance that has been less studied. As such, we fill a gap by finding that people's cognitive and behavioral responses toward personalized advertising can be explained by the extent to which they feel surveilled by advertisers and other corporations (Segijn, Opree, and van Ooijen 2022). Moreover, our research extends the DEAL framework by including surveillance responses that speak to policy surrounding dataveillance (i.e., desire for legal protection, interest in ad transparency information, and intended future use of ad transparency tools) in terms of imposing regulations to make ad targeting more transparent to consumers or mandating social media platforms to include transparency tools. These are not among outcome variables included in prior work on the DEAL framework to date, but they do have social importance because research on how ad transparency tools affect consumer attitudes can inform policymaking.

Boundary Conditions

The final goal of this research was to test privacy concern and privacy cynicism as boundary conditions for the effects of perceived surveillance on surveillance responses, as suggested by the DEAL framework. Results of several moderation analyses were not as expected but tell an interesting story. First, the

findings show that privacy concern served as moderator in many tests, whereas privacy cynicism did not. Second, the predictive power of perceived surveillance on perceived risks, desire for regulation, future use of privacy protection strategies and ad transparency tools was even stronger among less concerned individuals. It could be that highly concerned individuals had high baseline perceived surveillance; thus, concern did not increase the already-high relationships between perceived surveillance and the various surveillance responses. Other findings aligned with our expectations, for example, the strong negative relationship between perceived surveillance and perceived benefits among highly concerned individuals, and a strong positive relationship between perceived surveillance and negative affect among highly concerned individuals.

Privacy cynicism did not significantly moderate any relationships except for one in which it strengthened the expected positive relationship between perceived surveillance and future use of ad transparency tools. Privacy-cynical individuals likely considered ad transparency tools as a potentially effective means to help them combat privacy threats because they may have lost confidence in the effectiveness of other privacy protection mechanisms. Overall, the results concerning boundary conditions provide a more nuanced understanding of for whom perceived surveillance may be more likely to translate into actual surveillance responses.

Limitations and Future Research

Our study provides new data on public attitudes toward and usage of available ad transparency efforts and answers the call for testing key relationships proposed by the DEAL framework (Strycharz and Segijn 2022). However, it is exploratory in nature and thus not an exhaustive test of the model. Future research should also test other relationships proposed by the DEAL framework; for example: How do perceived surveillance and surveillance responses impact responses to specific advertisements? Do perceived surveillance and surveillance responses differ by other individual differences and contextual factors, such as demographics, persuasion knowledge, or perceived intentions of personalized advertising?

Our finding that varying the extensiveness of ad transparency led to no difference in online perceived surveillance—which is interesting in itself—calls for future studies to investigate what *kinds* of surveillance episodes impact surveillance perceptions. Indeed, while our operationalization of levels of ad transparency information centered around its extensiveness, future research is needed from a message framing and design perspective to investigate *how* to best inform users about advertiser data collection and targeting practices, as suggested by Strycharz and Segijn (2022). It may be that the extensiveness or type of ad transparency information matters less than the way that information is disclosed to consumers.

Another limitation is that the experimental stimuli were presented in the specific context of Instagram, whereas the perceived surveillance measure assessed people's feelings of being watched on social media and the internet generally (which was necessary to answer Research Question 1), and thus participants may have reported their general baseline perceived surveillance rather than judging it based on seeing the stimuli. Consequently, our study finds that seeing different ad transparency information on a particular platform (i.e., Instagram) does not change people's general perceived surveillance. Future research should replicate this finding by measuring perceived surveillance by the specific platform or advertiser shown in the stimuli.

Whereas many of the measures used in this study were based on existing scales, some were developed specifically for this study; further psychometric evaluation of these measures is thus needed. One of our measures in particular, general privacy concern, had somewhat low reliability, although not terribly different from past research (see Dienlin and Metzger 2016). It is important that future research confirm our results using additional measures of privacy concern.

In addition, our U.S. sample is not representative of all online consumers. For example, greater concern about and legal protection of online consumer privacy in the European Union, as reflected in policies such as the General Data Protection Regulation (GDPR), may prompt different responses to personalized advertising and to ad transparency efforts for people who live outside the United States. Future research should continue to validate and elaborate the DEAL framework by testing cultural differences as a potential contextual boundary condition. Additionally, other policy-related measures, such as consumer empowerment and trust in the data economy as a result of exposure to ad transparency information, should be studied.

Finally, further research is needed to see if results from our experiment can be replicated using more naturalistic ad transparency stimuli. We showed participants in the treatment conditions an ad that was personalized for a hypothetical person as the

surveillance episode. Showing people an instance of a surveillance episode for a stranger rather than for themselves could have dampened some of our results. Indeed, stronger relationships might appear using a more personalized surveillance episode. That said, our design is similar to many others in this research literature (e.g., Dogruel 2019; Kim, Barasz, and John 2019; Meier, Oeldorf-Hirsch, and Krämer 2024).

In any case, our hope is that this study inspires further research that focuses not only on responses to personalized advertising, but also specifically to advertising transparency. This point is critically important to ongoing efforts to improve advertising transparency on both the regulation and software development sides (Edelson et al. 2021). Our main findings on this front, including that people are not very knowledgeable or satisfied with existing tools, that seeing more extensive ad transparency information helps people feel more knowledgeable and self-efficacious about transparency tools, and that most people feel surveilled by advertisers, which fuels their desire for both legal protection from targeting and for using transparency tools to understand how targeting works, contribute meaningfully to such efforts.

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